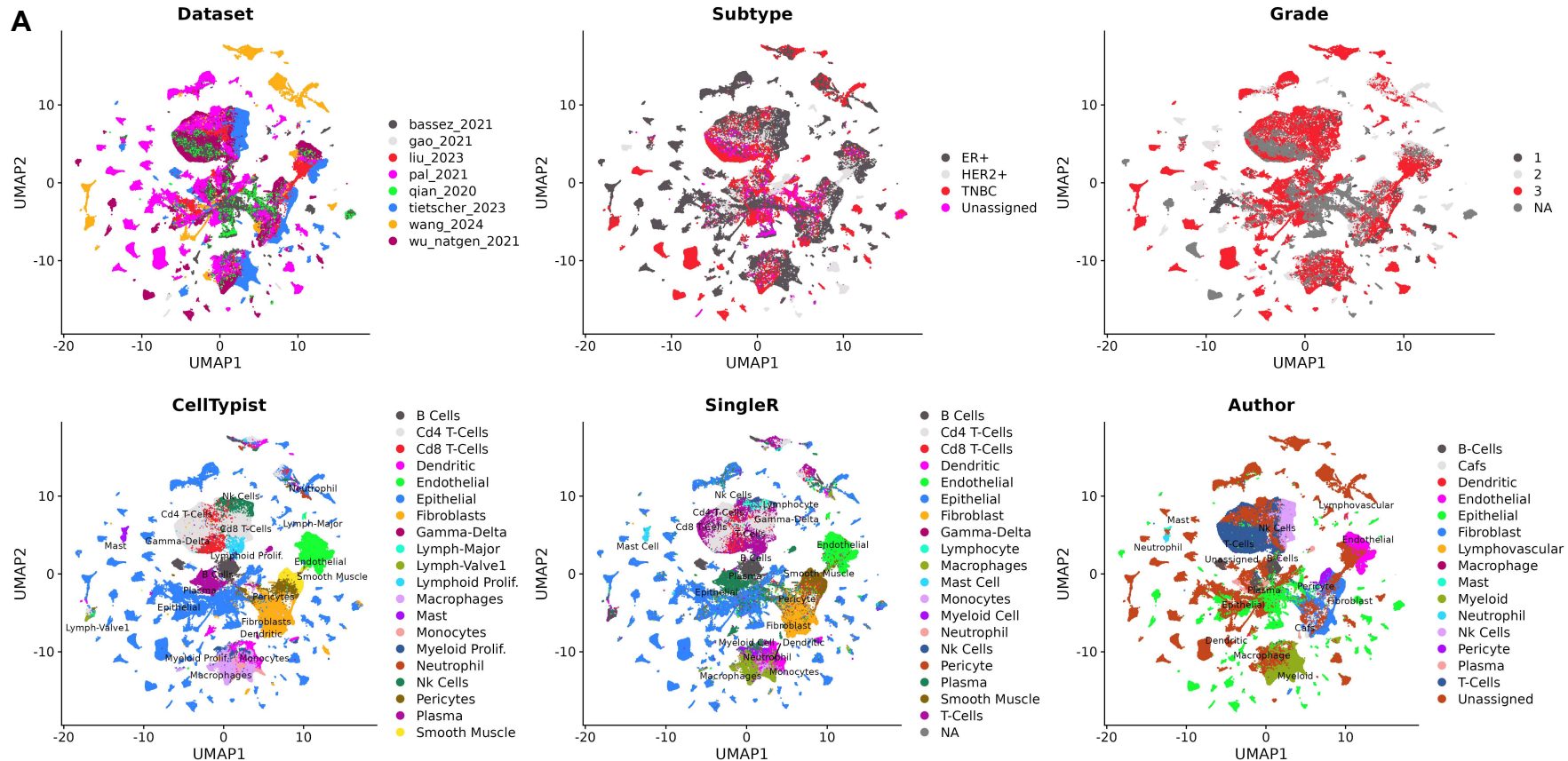
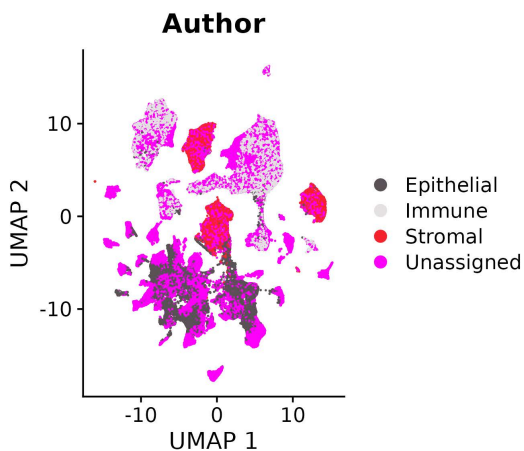
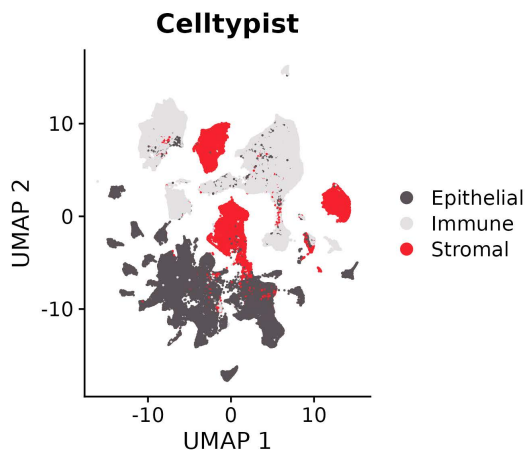
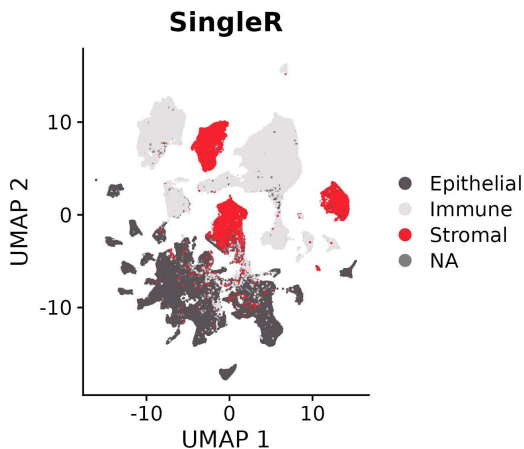
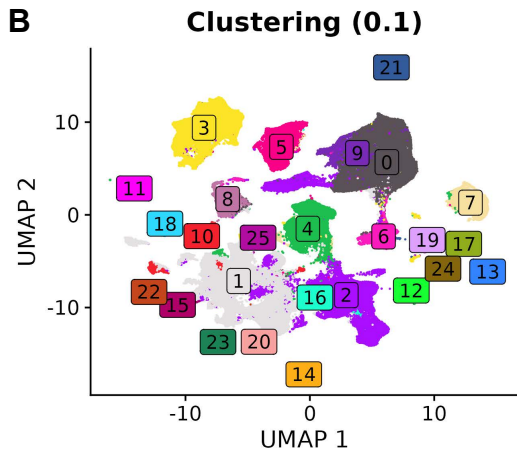
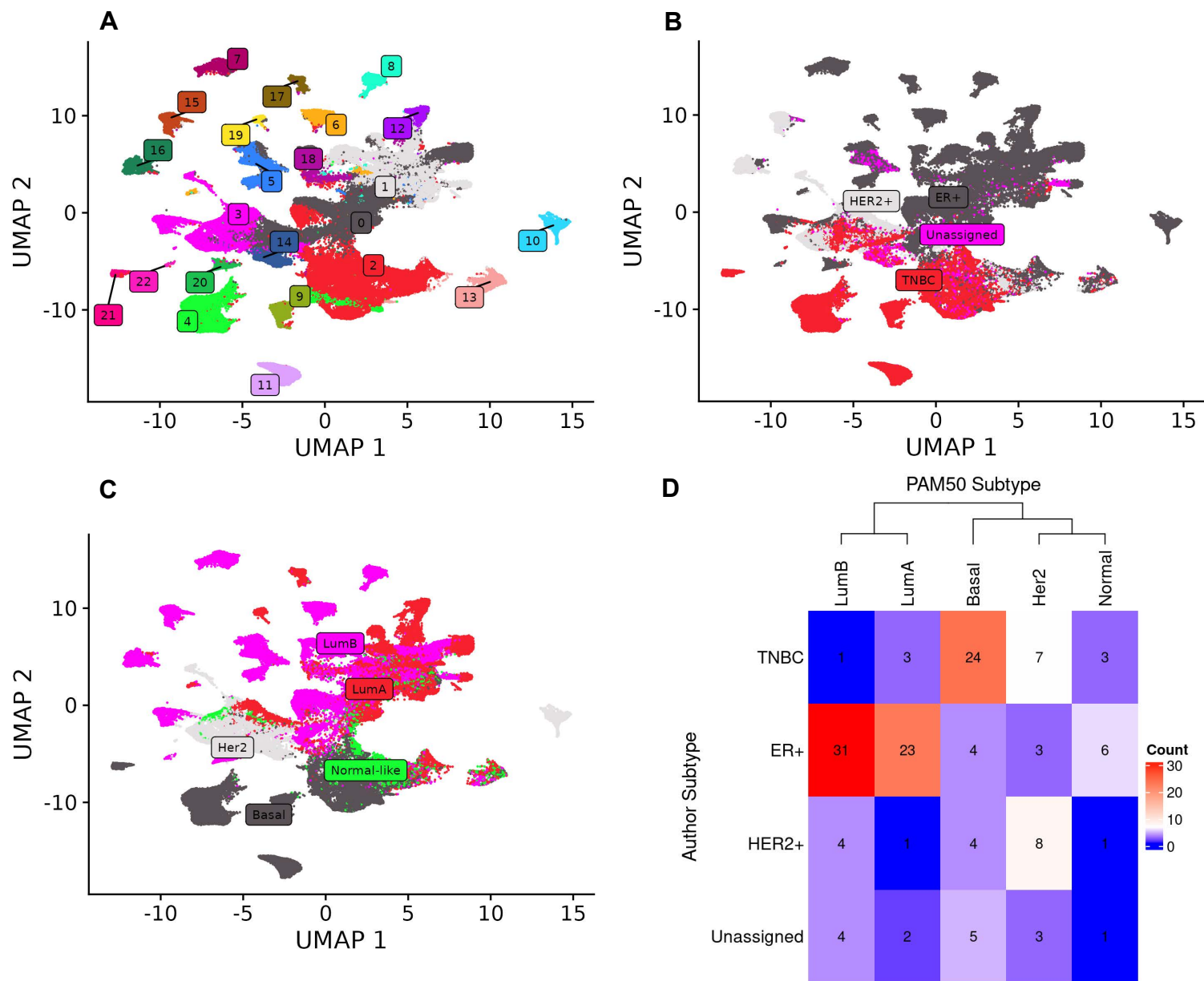




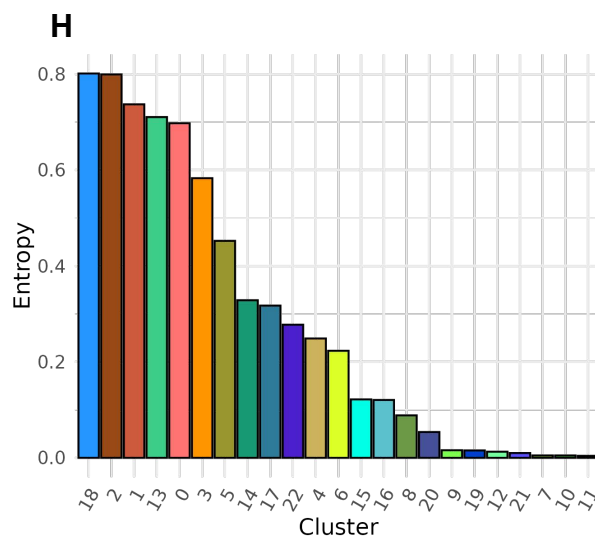
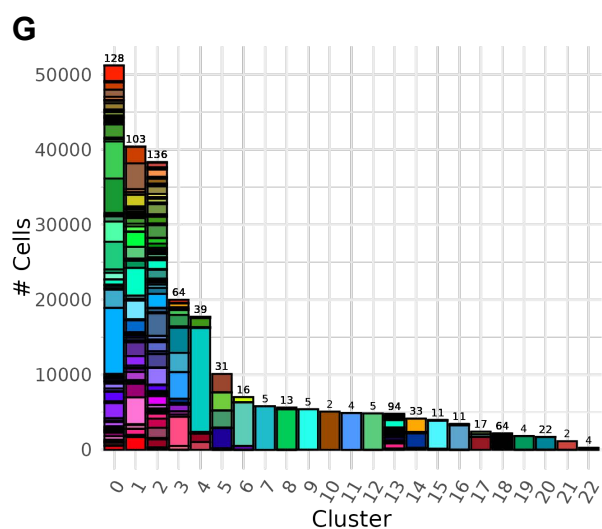
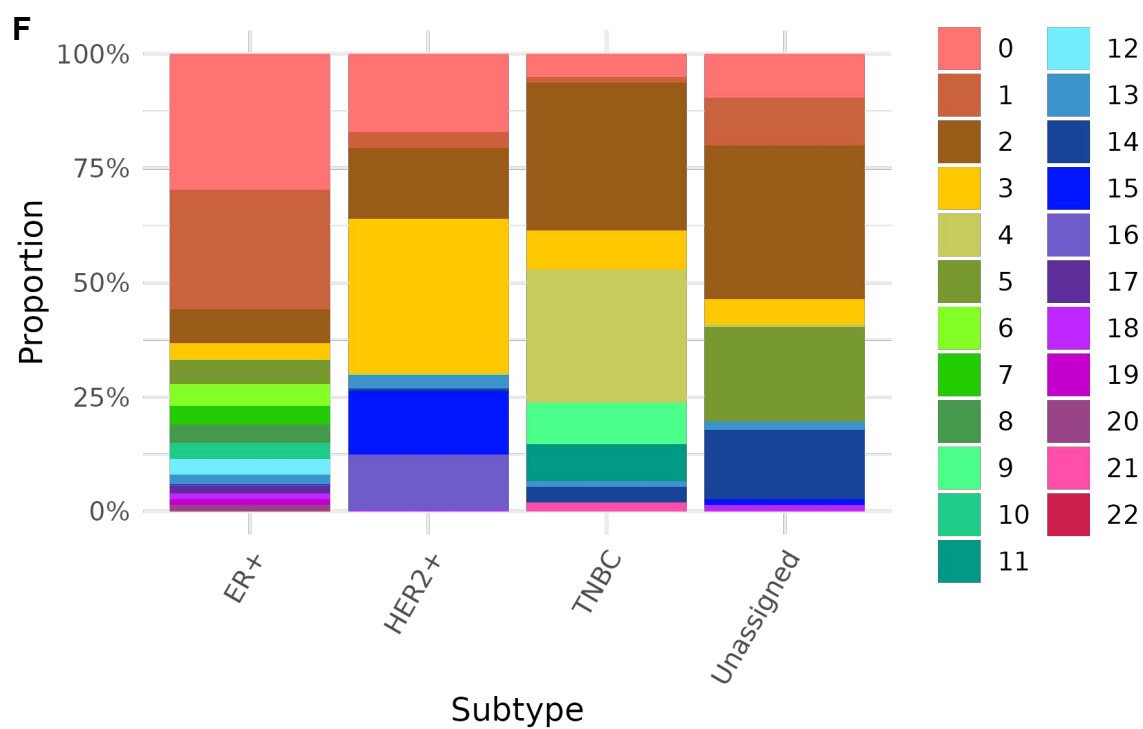
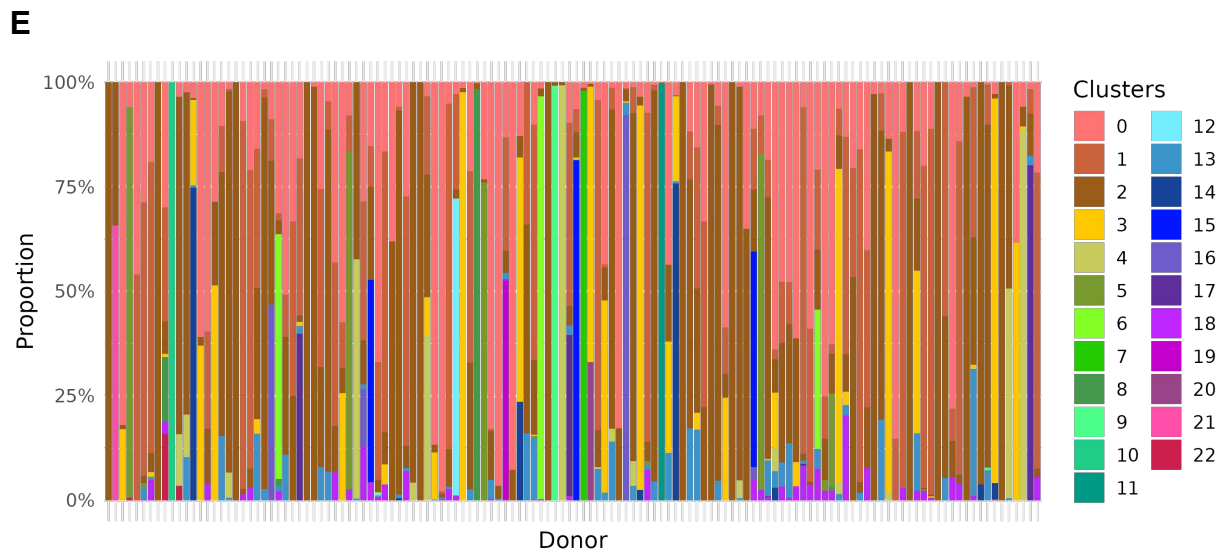
S1 - A) UMAP of unintegrated data colored by dataset, subtype, grade, cell typist, singleR, and author annotation.



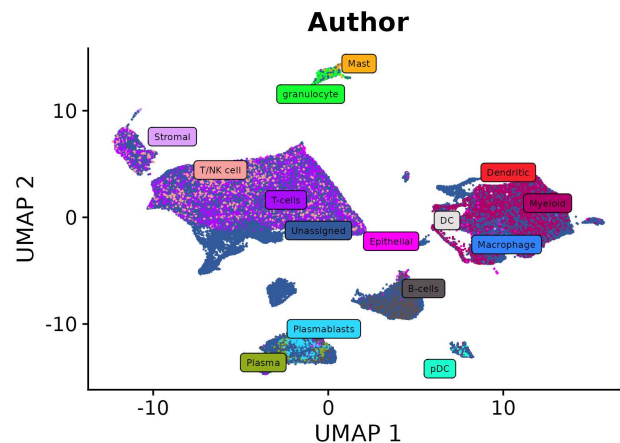
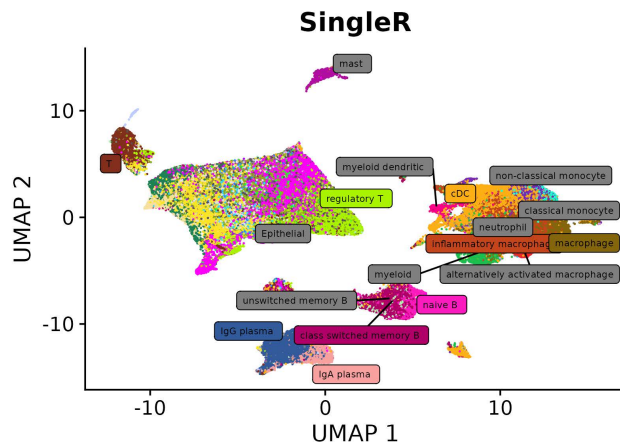
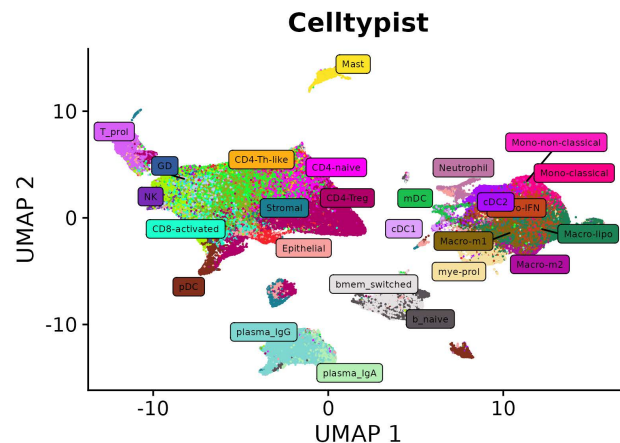
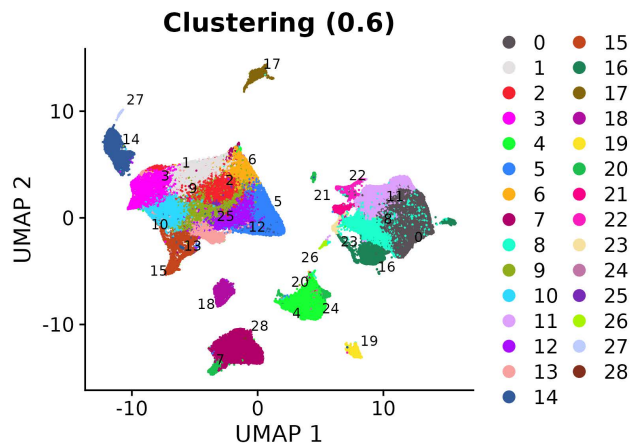
S1 - B) UMAP of RPCA integrated data colored by clustering, and broad cell type annotations from singleR, CellTypist, and author provided metadata.



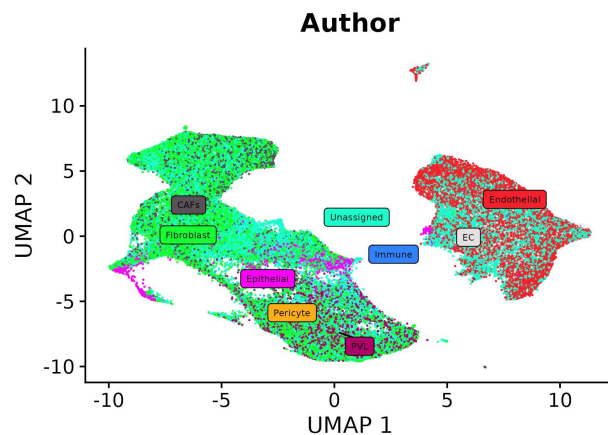
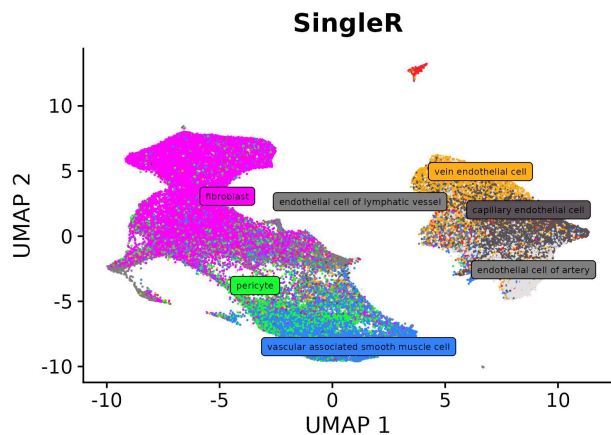
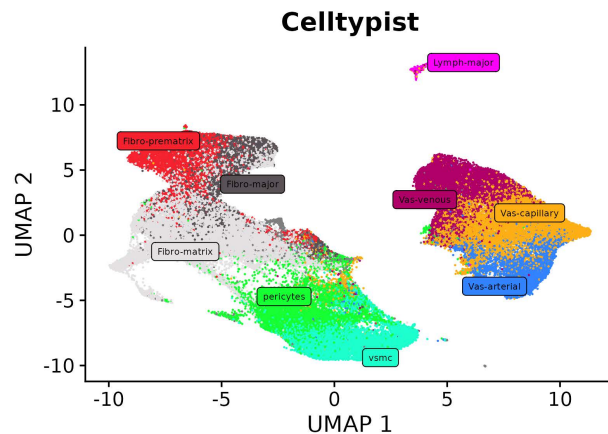
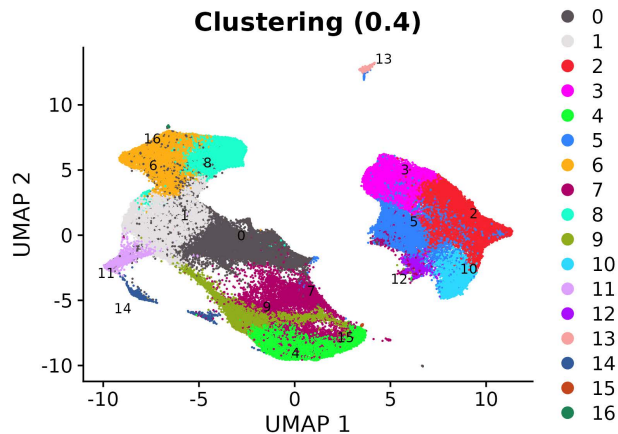
S2 - Cancer Epithelial Compartment Diversity. A) UMAP of cancer epithelial cells colored by clustering performed at 0.2 resolution. B) UMAP of cancer epithelial cells colored by tumor subtype. C) UMAP of cancer epithelial cells colored by PAM 50 subtype. D) Confusion matrix of intrinsic molecular subtype classifications (PAM50) and provided clinical subtype classifications



S2 - E) Distribution of epithelial clusters across each donor. **F)** Distribution of epithelial clusters across tumor subtypes. **G)** Distribution of patients across each of the clusters. **H)** Entropy of donor proportions across each of the clusters.

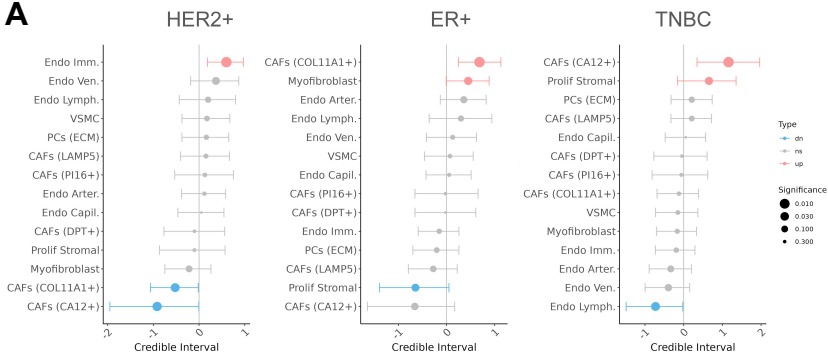


S3 - UMAP of immune compartment colored by clustering, and cell type annotations from singleR, CellTypist, and author provided metadata.

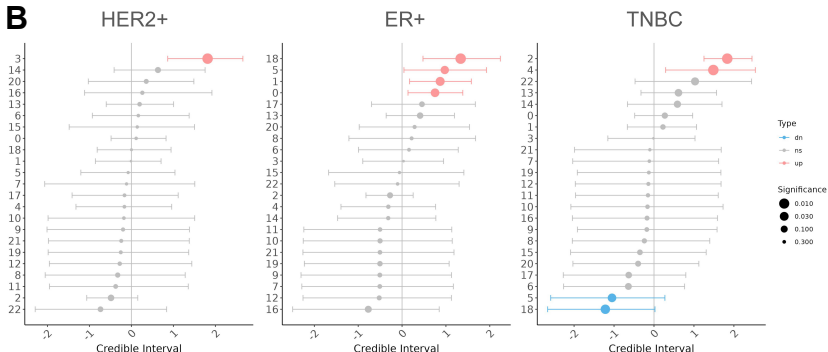


S4 - UMAP of stromal compartment colored by clustering, and cell type annotations from singleR, CellTypist, and author provided metadata.

A



B



S5 - Differential cell type estimates within the stromal compartment between HER2+, ER+ and TNBC tumors.
 B) Differential epithelial cell clusters between HER2+, ER+ and TNBC tumors.